

CS4733/5733: Computer vision for Autonomous Vehicles

Fall 2025

Final Exam – Deadline: December 21, 2025, 11:59 PM

Part 2: Theory Questions [30 points]

Question 1 [10 points: 3,3,2]: Assume three drones D1 and D2 and D3 are following different trajectories: a circle in x-y plane, a circle in x-z plane and an arc of parabola in y-z plane respectively. In the scene, we have highlighted the following elements:

- The world frame (x_w, y_w, z_w)
- The D1 body frame, centered in O1 with axes (x_1, y_1, z_1)
- The D2 body frame, centered in O2 with axes (x_2, y_2, z_2)
- The D3 body frame, centered in O3 with axes (x_3, y_3, z_3)

Positions

In the world frame, D1 and D2 and D3's origins are given by:

- $O_1^w = [\cos(t), \sin(t), 0]^T$
- $O_2^w = \left[\frac{1}{2}\cos(t), 0, \frac{1}{2}\sin(t)\right]^T$
- $O_3^w = [0, \cos(t), \cos(2t)]^T$

where t denotes time.

Orientation

- D1 reference frame is such that y_1 stays tangent to D1's trajectory for all time t and z_1 is parallel to z_w for all t (i.e., equivalently, roll = pitch = 0, yaw = t)
- D2 reference frame is such that x_2 stays tangent to D2's trajectory for all time t and y_2 is parallel to y_w for all t (i.e., equivalently, roll = pitch = t , yaw = 0)
- D3's reference frame moves with pure translation and we can assume that its axes are parallel to the world axes for all times t .

Question 1-1) In the problem formulation, we mentioned that D3's trajectory is an arc of parabola in the y-z plane of the world frame. Can you prove this statement?

- Hint: Parabola has a general form of $f(y) = ay^2 + by + c$ and note that $\cos(2t)$ can be written as: $\cos(2t) = 1 - 2\sin^2(t)$

Question 1-2) Compute o_2^1 , i.e., the position of D2 relative to D1's body frame as a function of t . Hint: write down the homogeneous transformations and compose them accordingly.

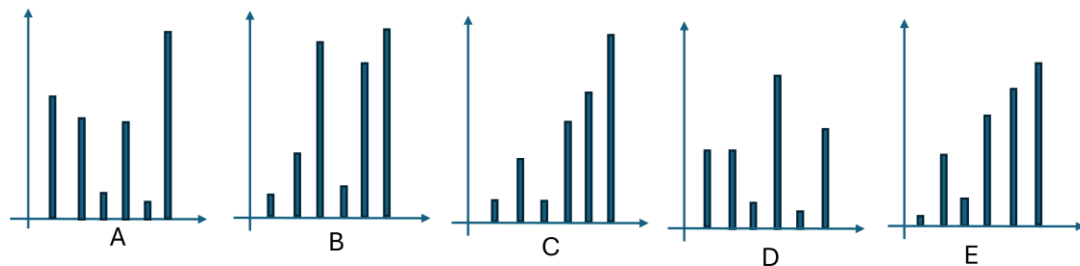
Question 1-3) Compute o_3^1 and show that o_3^1 describes a planar curve and find the equation of its plane. Hint: find a linear relation between z_3^1 and x_3^1 .

Question 2 [10 points: 5,5]

2-1) Assume we use an RGB-D camera (camera with depth information).

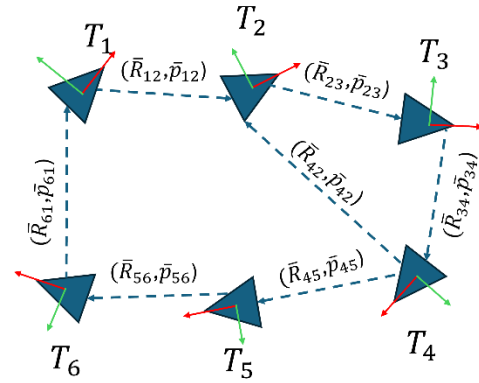
which has a built in IMU. What type of PNP algorithm (e.g., 8-point algorithm) we discussed in the class would you consider for estimating the camera pose? Why? Does the result of this algorithm suffer scale ambiguity? Explain why or why not?

2-2) Consider The bag of visual words histogram for 5 images as below. Which pair/s do represent the same place? Explain why?



Question 3 [10points: 1, 3, 3, 3]

Consider the pose graph shown on the right:



3-1) Circle or specify the loop closures.

3-2) Draw the Jacobian matrix and Hessian matrix of this graph. Showing the pattern as discussed in the class is enough.

3-3) In the same graph, we want to marginalize pose x_4 (i.e., T_4) What is the Jacobian matrix and Hessian matrix after this marginalization?

3-4) What is the Pose Graph after this marginalization?